

User Manual

Rail Delivery Group — Service Assurance
Version 1.4 | June 2026

1. Introduction

The **FRACC TIS Accreditation Status Report Generator** is a command-line Python tool that produces the monthly FRACC paper and supporting Excel data file automatically. It takes two user-supplied inputs — a collated Excel report and a Word template — and applies all required data, formatting, and document patches in a single run.

The tool was designed and refined by the Service Assurance team at Rail Delivery Group to replicate and replace the manual steps previously performed via Excel macros and manual copy-paste into Word.

1.1 Outputs

File	Description
FRACC - TIS Accreditation Status - Latest - <timestamp>.xlsx	Collated Excel with data grouped by Owning Group, TOC, System, and accreditation status
FRACC Paper - Accreditation Status Update - <date>.docx	Fully populated Word paper with updated dates, embedded chart, hyperlinked appendices
pie_chart_fracc.png	Accreditation status pie chart (intermediate file, embedded in the Word doc)

2. Prerequisites

2.1 Python version

Python 3.9 or later is required.

2.2 Install dependencies

Run the following command once to install all required packages:

```
pip install pandas openpyxl python-docx matplotlib lxml Pillow
```

2.3 Required input files

File	Format	Description
Collated TOC TIS Report	.xlsx	Master collated report produced by the TIS Accreditation Status Agent, or a direct export from the TIS system
FRACC Paper Template	.docx	The Word template from the previous month's paper. Must contain bookmarks, placeholders, and a table of contents

3. Running the Tool

3.1 Basic usage

Open a terminal, navigate to the folder containing `generate_fracc_paper.py`, and run:

```
python generate_fracc_paper.py \ --collated "TOC TIS Accreditation Status - Collated Report - v3.0.xlsx" \ --template "FRACC Paper Template.docx" \ --paper-date "16 June 2026" \ --output-dir "./output"
```

3.2 All parameters

Parameter	Required	Description / Example
<code>--collated</code>	Yes	Path to the collated Excel report. "TOC TIS Accreditation Status - Collated Report - v3.0.xlsx"
<code>--template</code>	Yes	Path to the Word template. "FRACC Paper Template.docx"
<code>--paper-date</code>	Yes	Paper date for the document header. "16 June 2026" or "16/06/2026"
<code>--output-dir</code>	No	Folder to write output files into. Default: current directory
<code>--news-image</code>	No	Path to an optional ASSIST News widget screenshot (.png). If supplied, replaces the news image placeholder in the Word doc

3.3 Expected console output

When the tool runs successfully you will see output similar to the following:

```
===== FRACC TIS Accreditation
Status Report Generator =====
Collated report : TOC TIS Accreditation Status...xlsx Template : FRACC Paper
Template.docx Paper date : 16 June 2026 Output dir : ./output
===== [1] Loading collated
report: ... [2] Applying peak-count logic... 182 rows → 121 unique combinations after
peak-count [3] Building FRACC Excel file... Saved FRACC Excel → FRACC - TIS
Accreditation Status - Latest - 20260616_0930.xlsx [4] Generating pie chart... Saved
pie chart → pie_chart_fracc.png [5] Patching Word document... ✓ Meeting Date → 'XX
Month' (highlighted) ✓ Paper Date → '16 June 2026' ✓ Update para date → '16 June 2026'
✓ Trailer table date → 'XX Month' (highlighted) ✓ Converted 16 appendix entries to
internal hyperlinks Saved Word document → FRACC Paper - Accreditation Status Update -
16 June 2026.docx ===== ✓
Generation complete =====
```

4. How It Works

4.1 Step 1 — Load & normalise

The tool reads the collated report and auto-detects its format. Two formats are supported:

Format	Detection	Notes
Pivot export	Has an 'Owning Group' column	Standard output from the TIS Accreditation Status Agent. Used directly.
Raw weekly format	Header at row 4, data from row 5	Direct export from the RSP portal. Owning Group is inferred where possible.

4.2 Step 2 — Peak-count logic

For every unique combination of **Owning Group + TOC/LPC + System + Machine Type ID + Version**, the tool selects the **highest device count ever recorded** across all historical weekly blocks in the collated file. The State and CoA Expiry date are taken from the same row as the peak count.

This approach avoids spurious dips caused by portal shifts being missing or incomplete on any given week, giving a stable and accurate representation of the true accredited fleet size.

4.3 Step 3 — FRACC Excel

A three-sheet Excel workbook is built:

Sheet	Content
TIS Accreditation Status	Data grouped by Owning Group header rows (RDG navy) → TOC sub-headers (light blue) → column
Pivot Data	Flat table of all peak-count records. Suitable for pivot tables, mail merge, or further analysis.
Accreditation Status Chart	Summary counts by State used to drive the pie chart. Shows total device count.

4.4 Step 4 — Pie chart

A Matplotlib pie chart is generated at 180 DPI showing the proportion of devices in each accreditation state. Labels showing percentage and device count are placed outside the wedges with leader lines. The legend is positioned to the right with no overlap.

4.5 Step 5 — Word document patches

#	What changes	Detail
1	Meeting Date	Set to 'XX Month' with yellow highlight. The reviewer fills this in before distribution.
2	Paper Date	Set to the value supplied via --paper-date.
3	Update paragraph	The bold date in 'this report was run on [date]' is updated to match the Paper Date.
4	Trailer table	The date cell in the Governance Lifecycle Tracking Grid (last table in the doc) is set to 'XX Month' with
5	Pie chart image	The existing chart placeholder (image1.png in the docx zip) is replaced with the newly generated cha
6	News image	If --news-image is supplied, the ASSIST News screenshot placeholder (image2.png) is replaced with
7	Appendix hyperlinks	All 16 entries in the appendix contents list (A–P) are converted to internal hyperlinks (w:hyperlink with

5. Word Template Requirements

The Word template must meet the following structural requirements for the patches to apply correctly. The easiest way to satisfy all requirements is to use the previous month's paper as the template.

Requirement	Details
Meeting Date paragraph	A paragraph beginning 'Meeting Date:' with a tab and the previous date as a run.
Paper Date paragraph	A paragraph beginning 'Paper Date:' with the previous date as a run.
Update paragraph	A paragraph containing the phrase 'this report was run on' with the date in two consecutive bold runs
Trailer/governance table	The last table in the document must have the meeting date in column 2 (index 1), row 1 (index 0), high
Image placeholders	Two embedded PNG images: image1.png (pie chart) and image2.png (news screenshot). These are r
Appendix bookmarks	Section heading paragraphs for each appendix must carry a Word bookmarkStart with names: Appen
Appendix contents list	Paragraphs before END OF PAPER reading 'Appendix A - ...' through 'Appendix P - ...'. These are co
Hyperlink character style	A character style named 'Hyperlink' must be defined (present in all Word documents by default).

6. Colour Reference

6.1 Accreditation state colours (Excel)

State	Cell Fill	Font Colour
Accredited	#C6EFCE (light green)	#276221 (dark green)
Pilot phase	#E2EFDA (very light green)	#375623 (forest green)
Accreditation expired	#FFC7CE (light red)	#9C0006 (dark red)
Application acknowledged	#FFEB9C (light amber)	#9C6500 (dark amber)

6.2 Excel structure colours

Element	Fill	Font
Owning Group header row	#1F3864 (RDG navy)	White, bold, 12pt
TOC sub-header row	#D6E4F0 (light blue)	#1F3864, bold, 10pt
Column headers	#EBF3FB (very light blue)	#1F3864, bold, 9pt
Data rows	White / alternating light grey	Black, 9pt

7. Appendix Structure

Appendix	Owning Group / Content	Bookmark Name
A	Go Ahead	Appendix_A
B	Transport UK Group (formerly Abellio Group)	Appendix_B
C	Serco / Transport UK Group (formerly Abellio Group)	Appendix_C

D	First Group	Appendix_D
E	Directly Operated Railway	Appendix_E
F	Arriva	Appendix_F
G	Transport for Wales	Appendix_G
H	London Overground	Appendix_H
I	GTS Elizabeth Line	Appendix_I
K	Heathrow Express Operating Company	Appendix_K
L	Scottish Rail Holdings	Appendix_L
M	Accreditation Status (summary + pie chart)	AppendixM
N	RDG Compliance Standards — Recent Updates	AppendixN
O	Third Party Retailer Systems	AppendixO
P	Governance Lifecycle Tracking Grid	AppendixP

Note: Appendix J is intentionally omitted. This is consistent with the original paper structure.

8. Troubleshooting

Symptom	Likely Cause	Fix
'Owning Group' column missing	Collated file is in raw weekly format, not pie chart format. This is handled automatically. Check that the file has a header row at row 4 with the following text: 'Owning Group'	
Meeting Date not updated	The template paragraph text does not contain a placeholder for the meeting date. Check the template paragraph text. The script searches for month names (January, February, etc.) and updates the date accordingly.	
Update para date not updated	The phrase 'this report was recorded on' is not present in the paragraph. Check the update section in the paragraph. Ensure the phrase is exact and the date is correct.	
Hyperlinks not appearing	Appendix list paragraphs do not contain the correct prefix. Check that the Appendix list paragraphs contain the correct prefix. The script matches on the prefix 'Appendix A' and replaces it with the correct prefix.	
'Appendix_' bookmark missing	Template was created before the script was updated. The script automatically adds the missing bookmark. If still failing, open the template and add the bookmark manually.	
Images not replaced	Template docx contains more images than the script can replace. The script replaces word/media/image1.png (if present) and word/media/image2.png (if present).	
ValueError: Cannot parse date string	Date string format not recognised. Use 'DD Month YYYY' (e.g. '16 June 2026') or 'DD/MM/YYYY'. The script falls back to the latter if the former fails.	
ModuleNotFoundError	Dependencies not installed	Run: pip install pandas openpyxl python-docx matplotlib lxml Pillow

9. File Naming Conventions

File	Name pattern	Example
FRACC Excel	FRACC - TIS Accreditation Status - Latest - YYYYMMDD	FRACC - TIS Accreditation Status - Latest - 20260616_0930.xlsm
FRACC Word paper	FRACC Paper - Accreditation Status Update - YYYYMMDD	FRACC Paper - Accreditation Status Update - 16 June 2026.docx
Pie chart	pie_chart_fracc.png	pie_chart_fracc.png (kept in output dir for reference)

10. Version History

Version	Date	Changes
1.0	May 2026	Initial release. Full pipeline: collated report → peak counts → Excel → pie chart → Word patches → h
1.1	June 2026	Added support for raw stacked collated format (multiple weekly blocks in a single sheet). Definitive LP
1.2	June 2026	Pixel-perfect Excel formatting to match reference file: Aptos font throughout, FFRRGGBB fill colours,
1.3	June 2026	Added native openpyxl PieChart to the Accreditation Status Chart sheet. Chart is linked directly to dat
1.4	June 2026	Removed 'Appendix X — ' prefix from Excel Owning Group header rows and from Word document se